Inequalities in healthcare utilisation in people with COVID-19, influenza and RSV infection

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GitHub repository: https://github.com/opensafely/winter_pressures_inequalities

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1. Background

Respiratory virus infections demonstrate health inequalities and disproportionately affect people living in deprived areas and people from ethnically minoritised backgrounds (1), this has been highlighted during the COVID-19 pandemic (2). The reasons for these differences are not clear, but are likely intersectional (3). Few studies have investigated the impacts of respiratory virus infections by both ethnicity and deprivation (4). A UK study found that deprivation was a mediator between ethnicity and COVID-19 infection, but used a simple model that did not take into account other factors such as comorbidity (5).

Inequalities in healthcare access have been targeted by government policy in the UK, (6) but still exist (7). Primary care staff shortages affect the most deprived areas disproportionately (8). People living in more deprived areas can find it difficult to get a primary care appointment (9) and are more likely to attend emergency departments (10,11). The way in which primary care accessibility can affect attendance of emergency departments is not well understood (11). There is excess multimorbidity in those living in areas of deprivation and people from certain ethnic backgrounds (12,13). A study of emergency attendances in London found that multimorbidity was the strongest predictor of emergency department use (12).

Ethnic inequalities are not independent and work through a variety of factors such as deprivation, migrant status, comorbidity and rurality. People living in areas of higher deprivation and from Pakistani and Black ethnic backgrounds were roughly 2 times more likely to be hospitalised with COVID-19 and influenza in England (14). Access to primary care could impact hospital admissions in different ways: people living in areas with higher deprivation and ethnic minorities may have higher multimorbidity and consult with their GP more frequently and become sicker with infections resulting in hospitalisation. Alternatively people who have difficulty in accessing primary care may consult less often and not receive the treatment they need, resulting in an increased risk of hospital admission.

The aim of this study is to measure whether there are ethnic and socioeconomic differences in healthcare utilisation for COVID-19, respiratory syncytial virus (RSV) and influenza, and to assess mediators of these differences.

2. Objectives

2.1. Primary Objectives

a. To estimate ethnic differences associated with healthcare utilisation for 1) COVID-19, 2) influenza, 3) RSV.

b. To estimate socioeconomic differences associated with healthcare utilisation for 1) COVID-19, 2) influenza, 3) RSV.
c. To measure the extent to which ethnic differences are mediated by other factors such as deprivation, rurality and comorbidity.

3. Methods

3.1. OpenSAFELY platform

All data will be linked, stored and analysed securely using the OpenSAFELY platform, https://www.opensafely.org/, as part of the NHS England OpenSAFELY COVID-19 service. No GP data from patients who have registered a Type-1 Opt out with their GP surgery will be included in this study. Data will include pseudonymised data such as coded diagnoses, medications and physiological parameters. No free text data are included. All code will be shared openly for review and re-use under MIT open license https://github.com/opensafely/winter_pressures_inequalities/tree/main. Detailed pseudonymised patient data is potentially re-identifiable and therefore will not be shared. A detailed description of the OpenSAFELY platform is available in Nab et al. (15)

Access to the underlying identifiable and potentially re-identifiable pseudonymised EHR data is tightly governed by various legislative and regulatory frameworks, and restricted by best practice. The data in the NHS England OpenSAFELY COVID-19 service is drawn from General Practice data across England where TPP is the data processor.

TPP developers initiate an automated process to create pseudonymised records in the core OpenSAFELY database, which are copies of key structured data tables in the identifiable records. These pseudonymised records are linked onto key external data resources that have also been pseudonymised via SHA-512 one-way hashing of NHS numbers using a shared salt. University of Oxford, Bennett Institute for Applied Data Science developers and PIs, who hold contracts with NHS England, have access to the OpenSAFELY pseudonymised data tables to develop the OpenSAFELY tools.

These tools in turn enable researchers with OpenSAFELY data access agreements to write and execute code for data management and data analysis without direct access to the underlying raw pseudonymised patient data, and to review the outputs of this code. All code for the full data management pipeline — from raw data to completed results for this analysis — and for the OpenSAFELY platform as a whole is available for review at github.com/OpenSAFELY.

3.2. Study Design

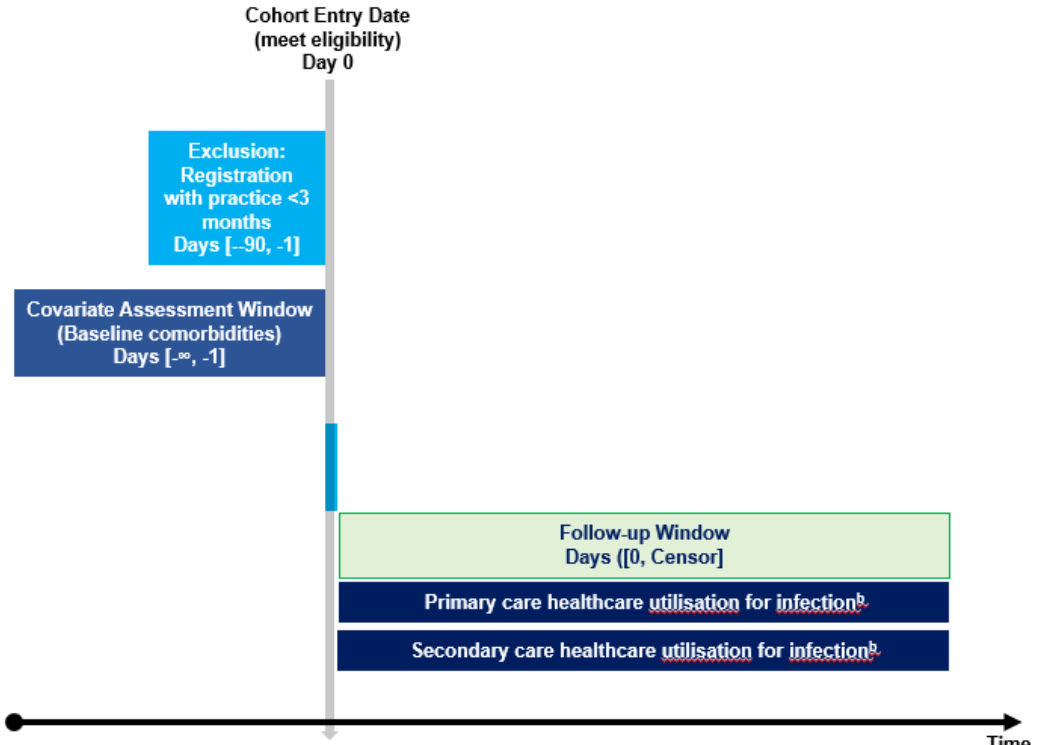
Study design: Cohort study.

The study periods will be different depending on the infection of interest:

- For COVID-19 the study period will be 1st August 2022 to current.
- For influenza and RSV the study period will be 1st September 2018 to current.

We will exclude the 2020/21 year; September 2020-August 2021 for influenza and RSV, and March 2020-February 2021 for COVID-19 as this year was unusual because of pandemic lockdowns.

Figure 1: Study design



[a] Censor at deregistration at practice, death or end of study period.
[b] Specific definitions of COVID-19, flu and RSV infection

Study periods:

- COVID-19: 1st August 2022 to current.
- For influenza and RSV: 1st September 2018 - current.

3.3. Data sources

3.3.1. Primary care data - TPP

SystmOne is a primary care clinical information system run by TPP, used by roughly one third of GP practices in England, with records for approximately 44% of the English population (16). It captures details about a patient that are electronically recorded or accessed by GPs, including symptoms, investigations, test results, diagnoses, prescriptions, demographic and social characteristics, etc.

3.3.2. ONS registered deaths

Date and cause of death based on information recorded when deaths are certified and registered in England and Wales. The death record includes the underlying cause of death

(i.e. the medical condition judged to be the underlying cause according to the rules of the *10th Revision of the International Classification of Diseases*) and up to 15 medical conditions mentioned on the death certificate.

3.3.3. Hospital admissions

Admitted Patient Care (APC) is the national data set for hospital admissions. Admitted Patient Care Spells (APCS) is part of Hospital Episode Statistics (HES) and is provided to OpenSAFELY via NHS Digital's Secondary Use Service (SUS). It includes admission and discharge dates, routes of admission, reason for admission, diagnoses treatments, and discharge destination.

3.3.4. Emergency attendances

The Emergency Care Data Set (ECDS) is the national data set for urgent and emergency care. ECDS is part of HES and is provided to OpenSAFELY via NHS Digital's SUS. It includes attendance, admission and discharge dates, locations, presenting complaints, and diagnoses.

3.4. Study Population

3.4.1. Inclusion / Exclusion Criteria

Study populations: The inclusion and exclusion criteria will be assessed on 1st March (COVID-19) or 1st September (influenza/RSV) each year through the study period.

Inclusion criteria:

- Registration at practice for at least three months on 1st September in year of interest, unless age less than 1 where there will be no restriction, allowing babies to enter the cohort when aged less than 3 months.

Exclusion criteria:

- People >110 years of age due to risk of incorrectly recorded age;
- People with missing age or sex as an indicator of poor data quality.

The index date will be the 1st March/September of the first year the person is eligible. People will be censored at the end of registration at the practice, death or end of the study period.

3.5. Study Measures

3.5.1. Exposure

Ethnicity determined using SNOMED-CT codes and supplemented with ICD-10 codes from SUS data, categorised in the 2001 ONS Census groupings (17):

- White

    - Subcategories: 1) British, 2) Irish, 3) Any other White background.
- Asian
    - Subcategories: 1) Indian, 2) Pakistani, 3) Bangladeshi, 4) Any other Asian background.
- Black
    - Subcategories: 1) African, 2) Caribbean, 3) Any other Black background
- Mixed
    - Subcategories: 1) White and Caribbean, 2) White and African, 3) White and Asian, 4) Any other Mixed background
- Other
    - Subcategory: 1) Chinese, 2) Any other background
- Unknown

3.5.2. Outcomes

Primary or secondary care utilisation for the infection of interest: an appointment (primary care) or admission (secondary care) with SNOMED-CT code (primary care) or ICD-10 code (secondary care) specific to infection (COVID-19, influenza or RSV). As developed in the disparities in COVID-19, influenza and RSV project. Multiple outcomes will be allowed during follow-up, but we will have a washout period of one month after healthcare utilisation, to avoid over-inflating utilisation. Any infection utilisation events within the same month will be considered the same infection and will not be counted.

3.5.3. Covariates

The following covariates will be measured at index date (i.e. first entry into the study), unless specified otherwise:

- Age at index date
    - Categorised into 0-4, 5-17, 18-64, 65-74, 75-79, 80-84, 85+.
- Sex
- Rural-urban classification based on the patient's address.
    - 5 levels: urban major conurbation, urban minor conurbation, urban city and town, rural town, rural village
- Deprivation:
    - 5 categories (1 = most deprived to 5 = least deprived) based on the Index for multiple deprivation 2019 (IMD).
- Comorbidity at index date
    - Based on the Cambridge Multimorbidity score (CMS) (18) measured using SNOMED-CT codes prior to index date.
- High risk vaccination group, i.e. those who are eligible for free vaccination.
    - Based on Green book information, measured using SNOMED-CT codes prior to index date, time updated each year eligible.
- Vaccination status measured as follows:

    - Influenza:
        - Vaccinated in the year prior (assessed on 1st September each year)
        - Vaccinated in the current year (September-August) (time-updated)
    - COVID-19:
        - Vaccinated in the year prior (assessed on 1st March each year): this will be based on the most recent vaccination they were eligible for (Table 1).
        - Vaccinated in current year (March-February) (time-updated)
    - RSV
        - Ever vaccinated (time-updated)
- BMI
    - Most recent BMI (within the last 2 years) based on height and weight measurements or coded BMI.
- Migrant status at index date
    - Determined through SNOMED-CT codes (19).
- English language proficiency at index date
    - Determined through SNOMED-CT codes for language spoken and needing an interpreter.
- GP utilisation in year prior to index
    - Count of appointments with a GP and/ prescriptions, as this is likely to be skewed the count will be categorised as: zero, 1-5, 6-10, 11+ based on findings of a median of 8 GP appointments per person per year in 2019 (20)

For all covariates we will include an unknown category where the information is missing.

Table 1: COVID-19 vaccination eligibility timeline

	Eligible groups
Sept 2021	● Age >49 years ● Clinical risk groups* age 16-49 ● Care home residents ● Front line and social care workers[%] ● Carers age >16[%] ● Household contacts of immunosuppressed age >16[%]
November 2021	● Age >= 16 ● Clinical risk group or in contact with ICP/pregnant age 12-15
March-June 2022	● Age >=75 years ● Care home residents ● Immunosuppressed individuals over 12 years
1st September 2022	● Age >= 50, ● Care home residents and staff ● Immunosuppressed individuals ● Clinical risk groups age 5-49 years ● Close contact with immunosuppressed individuals age 5-49 years[%] ● Carers aged 16-49 years[%]

	• Front line health and social care[%]
1st March 2023	• Age >=75 as of 30th June 2023, • Care home residents aged >65 • Immunosuppressed individuals 5–74
1st September 2023	• Age >= 65 • Care home residents • Clinical risk groups aged 6 months to 64 years • Health and social care workers[%] • Close contact with immunosuppressed individuals and aged 12-64 years[%] • Carers aged 16-64 years[%]
1st April 2024	• Age >=75 as of 30 June 2024 • Care home residents >=65 • Immunosuppressed individuals aged 6 months to 74 were offered an additional dose from.
1st September 2024	• Age >= 65 • Care home residents • Clinical risk groups aged 6 months to 64 years
1st April 2025	• Age >=75 as of 30 June 2025 • Care home residents >=65 • Immunosuppressed individuals aged 6 months to 74 were offered an additional dose from.

* Clinical risk groups are defined in the COVID-19 green book.

[%] Unable to identify, as not consistently coded in EHR data.

Codelists:

Variable	**Primary care or secondary care**	**Codelist**
COVID-19	Primary care	SNOMED-CT
	Secondary care	ICD-10
Influenza	Primary care	SNOMED-CT
	Secondary care	ICD-10
RSV	Primary care	SNOMED-CT
	Secondary care	ICD-10
Ethnicity	Primary care	SNOMED-CT
Migrant status	Primary care	SNOMED-CT
English language proficiency	Primary care	SNOMED-CT

Cambridge comorbidity score	Primary care	Hypertension: SNOMED-CT Anxiety/depression: SNOMED-CT Painful condition (Osteoarthritis): TBD Hearing loss: SNOMED-CT Irritable bowel syndrome: SNOMED-CT Asthma:SNOMED-CT Diabetes mellitus: SNOMED-CT Coronary heart disease: SNOMED-CT Chronic kidney disease: SNOMED-CT Atrial fibrillation: SNOMED-CT Constipation: TBD Stroke/transient ischemic attack (TIA): SNOMED-CT COPD: SNOMED-CT Connective tissue disorder: SNOMED-CT Cancer: SNOMED-CT Alcohol problems: SNOMED-CT Heart failure: SNOMED-CT Dementia: SNOMED-CT Psychosis/bipolar disorder: SNOMED-CT Epilepsy: SNOMED-CT

3.6. Statistical Analysis

Descriptive statistics:

- Summarise covariates at index date using descriptive statistics, stratified by ethnic group.

Measuring ethnic differences in healthcare utilisation for infection:

- Each month we will determine the rate of primary care and secondary care healthcare utilisation (combined and each setting separately) per 100,000 people within each ethnic group. The rates will be plotted in a line graph.

- We will use a Cox proportional hazards regression model to measure ethnic differences in the hazard of first health care utilisation, with White ethnicity as the reference. Models will be stratified by year to allow for different patterns of infection within each season.

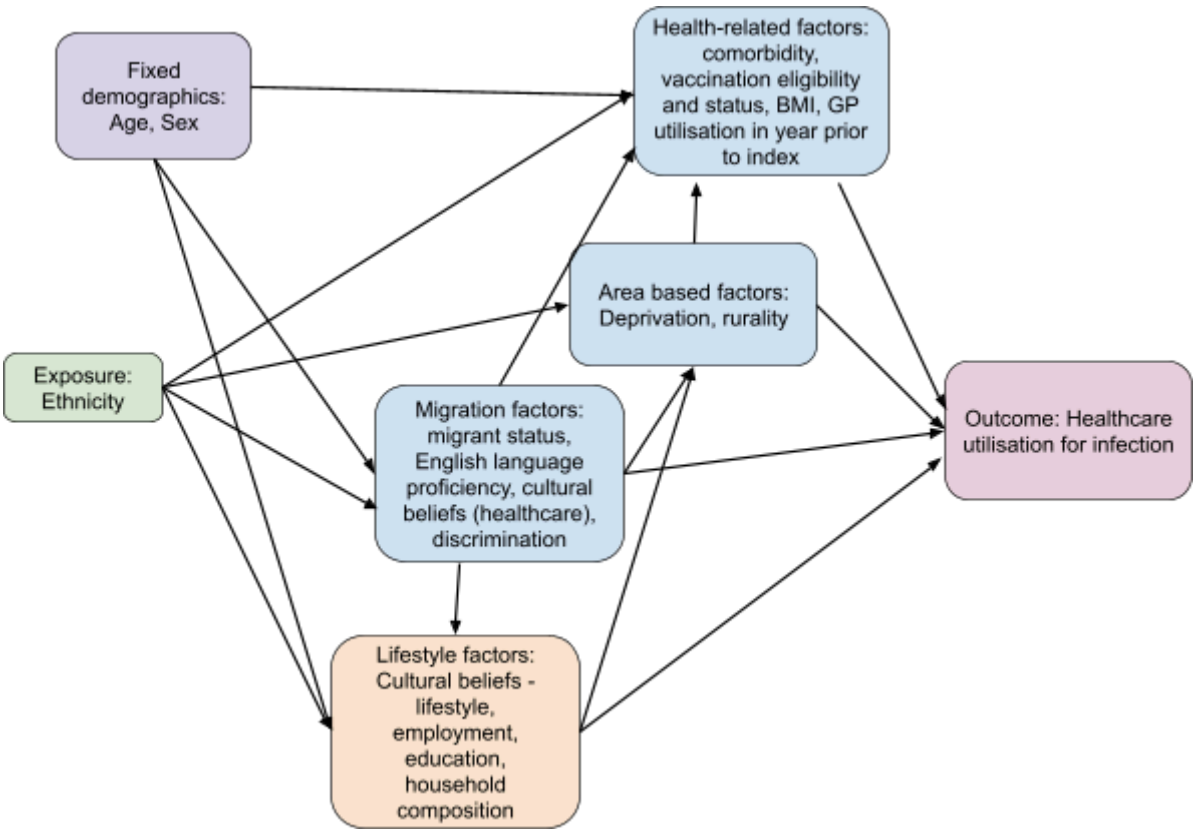
Mediation analysis:

Ethnic inequalities are not independent and work through a variety of factors such as deprivation, migrant status, comorbidity and rurality. We can use mediation analysis to estimate the direct and indirect effects to understand the proportion of ethnic differences mediated by other factors, which can ultimately help with prevention strategies (21,22). This type of analysis is appropriate for exploring ethnic differences as ethnicity is not modifiable, therefore there are no confounders (common causes) of ethnicity and healthcare utilisation for infections. However there are factors (mediators) on the pathway from ethnic differences to healthcare utilisation that are of interest and could potentially be intervened on (23).

Mediation analysis relies on several strong assumptions:

- No unmeasured exposure-outcome confounding
- No unmeasured mediator-outcome confounding
- No unmeasured exposure-mediator confounding
- No mediator-outcome confounder affected by the exposure.

Figure 2 is a Directed Acyclic Graph (DAG) depicting possible causal relationships between ethnicity, healthcare utilisation, fixed demographic factors and potential mediators. Although there are some factors within figure 2 that we are unable to measure, in particular some lifestyle characteristics, some aspects of these characteristics are captured within measured variables, such as age, migration factors.



Note: The lifestyle variables, cultural beliefs and discrimination are unable to be measured, or are unlikely to be well captured.

Causal mediation models estimate 1) the controlled direct effect of the exposure on the outcome (i.e. the effect not mediated, where the mediator is fixed to a specific value), 2) the natural direct effect of the exposure on the outcome (the effect of the exposure on the outcome not through the mediator), 3) natural indirect effect (the effect of the exposure on the outcome that operates through the mediator). The total effect can be decomposed into the natural direct and indirect effects (22). Such models depend on assumptions about causal relationships between the exposure, mediators, outcomes and other relevant variables. The applicability of specific methods to examine mediation is often dependent on the type of variables that the exposure, outcome and mediators are (continuous, categorical, binary).

Causal mediation methods are often used to examine the role of a specific mediator. In our study there are multiple mediators: this situation is more complex to analyse. There are several pathways to consider, as there are indirect effects for each mediator and potential interactions. There are methods to measure mediation of several variables jointly (24). We will investigate each of the measured variables listed under health related factors, area based factors and migration factors (Figure 2) as potential mediators. As a first exploratory step we will consider each of the mediators individually, we will then combine any that are correlated into one model. At this point we will decide whether using a multiple mediator model is appropriate and if so decide on the model to use.

To perform the mediation analysis we will fit: 1) a mediator model with exposure and covariates as independent variables, 2) an outcome model with the exposure, mediator and covariates as independent variables. We will then generate the average causal mediation effect and average direct effects using the mediate function in the R package “mediation” (25).

The types of models used are dependent on the mediator and outcome variable type. There are several types of variable for the mediators: binary (vaccination eligibility, migration status, English language proficiency, comorbidity) will use a logistic regression model, categorical (deprivation, rurality) will use multinomial logistic regression model and continuous (BMI) will use a linear regression model.The outcome is time-to-event, for this type of outcome the model used depends on how rare the outcome is (26,27), if the outcome is rare (<10%) we will use a Cox proportional hazards regression model, if the outcome is not rare we will use an accelerated failure model. We will use bootstrapping for 95% confidence intervals.

We plan to use a weighted approach to the causal mediation analysis, the aim of the weighting is to create a pseudo-population in which the exposure and mediators are independent. To generate inverse probability weights we will use a multinomial logistic regression model for ethnicity, including covariates (age, sex) and mediators (migration status, English language proficiency, IMD, rurality, comorbidity, vaccination status and BMI).

Mediation analyses will be conducted using the CMAverse R package.

3.6.1. Sensitivity Analyses

We will also conduct sensitivity analysis of the assumptions (detailed below) (28).

1. Use e-value methods to measure the strength an unmeasured confounder would need to be to explain direct or indirect effects.
2. Explore the impact of the infection definitions by using broad codelists to expand the infection definitions
3. Remove the inclusion criteria requiring 3 months registration prior to index date as we understand that past EHR for new patients are immediately available, therefore this restriction may not be necessary.

4. Limitations

- Causal mediation analysis relies on a number of assumptions, one of which is that there is no unmeasured confounding. It is possible that there is unmeasured confounding, however we will explore this through e-values.
- We do not know infection status, therefore we cannot estimate healthcare utilisation among infected people.
- Misclassification: as with all electronic health record studies diagnoses rely on clinical coding, therefore if events are not coded they may be misclassified. This is not an issue for the outcome (healthcare utilisation), but may affect comorbidities.

We are using codelists reviewed by people with clinical expertise, and will use validated definitions where available to minimise misclassification.

5. Disclosure Control

To manage the risk of disclosure–that is, the reidentification of individuals within the data–only aggregated results will be viewed and requested for release from the secure environment, and statistical disclosure control (SDC) will be applied to all results. This will include redaction of low values, rounding, and / or redesigning outputs to avoid sparse table cells, as required. To ensure SDC has been properly applied, all results will be reviewed by two trained members of the Bennett Institute prior to release from the secure environment.

6. Software and Reproducibility

Data management will be performed using Python, with analyses carried out using Python and R. Code for data management and analysis as well as codelists archived online [https://github.com/opensafely/winter_pressures_inequalities].

7. Patient and Public Involvement and Engagement

We will meet with the Patient and Public Involvement and Engagement (PPIE) advisory group, affiliated with the wider Winter pressures project, to provide feedback throughout the duration of this project. The group comprises people from across England with lived experience of accessing primary and secondary care services, including some members with multiple long-term conditions

OpenSAFELY has involved patients and the public in various ways: we developed a public website that provides a detailed description of the platform in language suitable for a lay audience (https://opensafely.org); we have participated in two citizen juries exploring public trust in OpenSAFELY; we have co-developed an explainer video (https://www.opensafely.org/about/**);** we have patient representation who are experts by experience on our OpenSAFELY Oversight Board; we have partnered with Understanding Patient Data to produce lay explainers on the importance of large datasets for research; we have presented at various online public engagement events to key communities (e.g., Healthcare Excellence Through Technology; Faculty of Clinical Informatics annual conference; NHS Assembly; HDRUK symposium); and more. To ensure the patient voice is represented, we are working closely to decide on language choices with appropriate medical research charities (e.g., Association of Medical Research Charities). We will share information and interpretation of our findings through press releases, social media channels, and plain language summaries.

8. Dissemination

We will disseminate findings through a peer-reviewed journal and conferences, following the RECORD guideline. We will also disseminate a lay summary through patients groups and blog posts.

9. Administrative

9.1. Timelines and approvals

NHS England will oversee the final approval for all publication ready papers, reports or presentations, to check that the outputs align with the stated application purpose.

We plan to have outputs in 2026.

9.2. Ethics

This work is part of the winter pressures project, the project has ethical approval from HRA and Health and Care Research Wales (HCRW) on 04 December 2024, REC reference: 24/HRA/5020.

9.3. Funding

This study/project is funded by the NIHR Health and Social Care Delivery Research programme (NIHR158218).

Funding information for the OpenSAFELY platform can be found here: https://www.opensafely.org/policies-for-researchers/#information-governance-ethics-and-funding-content-policy.

9.4. Information governance

Information governance for projects conducted in OpenSAFELY can be found here: https://www.opensafely.org/policies-for-researchers/#information-governance-ethics-and-funding-content-policy.

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